



## Brief Paper

# Payload trajectory tracking control for aerial transportation systems with cable length online optimization<sup>☆</sup>

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## ABSTRACT

Cable-suspended aerial transportation systems are employed extensively across various industries. The capability to flexibly adjust the relative position between the multirotor and the payload has spurred growing interest in the system equipped with variable-length cable, promising broader application potential. Compared to systems with fixed-length cables, introducing the variable-length cable adds a new degree of freedom. However, it also results in increased nonlinearity and more complex dynamic coupling among the multirotor, the cable and the payload, posing significant challenges in control design. This paper introduces a backstepping control strategy tailored for aerial transportation systems with variable-length cable, designed to precisely track the payload trajectory while dynamically adjusting cable length. Then, a cable length generator has been developed that achieves online optimization of the cable length while satisfying state constraints, thus balancing the multirotor's motion and cable length changes without the need for manual trajectory planning. The asymptotic stability of the closed-loop system is guaranteed through Lyapunov techniques and the growth restriction condition. Finally, simulation results confirm the efficacy of the proposed method in managing trajectory tracking and cable length adjustments effectively.

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## 1. Introduction

With the advancement of electronic technologies and control algorithms, unmanned aerial vehicles (UAVs) (Kong et al., 2025; Lv et al., 2022; Meng et al., 2025; Yang, Xian, et al., 2024; Yang, Yu, et al., 2024) have become essential tools across various industries, including agriculture, logistics, and rescue operations, etc. As one of the applications, using multirotor UAVs for suspended cargo transportation not only significantly enhances the efficiency of material delivery but also facilitates construction and rescue operations in environments inaccessible to ground vehicles. As a result, numerous studies have been conducted on cable-suspended aerial transportation systems in recent years.

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Conventionally, the primary research objective is to achieve effective payload delivery across various scenarios through control and planning methods. In certain situations, inducing large swing angles of the payload is desirable, particularly during obstacle avoidance or while maneuvering through narrow spaces (Tang et al., 2018; Wang et al., 2024; Yu et al., 2022). For instance, Tang et al. (2018) employ a downward-facing camera to estimate and capture the motion of the payload, facilitating rapid passage through slalom courses by generating large payload swing angles. Similarly, Yu et al. (2022), integrate constraints on cable direction into their trajectory generator, creating an aggressive payload swing trajectory that enables efficient window crossing. Wang et al. (2024) have developed an impact-aware planning and control framework that combines agile flight with hybrid motion modes, allowing the aerial transportation system to navigate through narrow circular gates. Conversely, for the purposes of transportation safety and payload protection, suppressing large payload swings is often necessary (Lee & Son, 2020; Xian et al., 2019; Yu et al., 2023). Xian et al. (2019) design a trajectory planning strategy that combines target positioning and anti-swing components to suppress payload swings without iterative optimizations. Analogously, Lee and Son (2020) introduce a dynamically feasible trajectory generator paired with an anti-swing tracking controller, which effectively dampens payload oscillations while allowing for transient aggressive motions. In addition,

a control framework that tracks a virtual point along the cable, effectively handling constant external disturbances and mitigating payload oscillations, is implemented by Yu et al. (2023). Furthermore, other studies have prioritized payload trajectory tracking over swing control (Cabecinhas et al., 2019; Kong et al., 2024). By using the full dynamics of the aerial transportation system, a backstepping control scheme is designed in Cabecinhas et al. (2019) to track the desired position of the point-mass payload. Besides, Kong et al. (2024) develop an adaptive backstepping control scheme for systems with unknown payload mass, setting pre-specified performance specifications for both payload position and cable direction.

However, most existing works focus on systems with fixed-length cables, which indicates that the payload's hoisting/lowering motion can only be achieved by changing the motion of the multirotor, making it difficult to complete certain tasks, such as navigating through narrow tunnels. Although some fixed-length cable systems can swing up the payload through narrow gaps using certain planning methods, it is not qualified for such tasks if the cave becomes longer. Moreover, fixed-length cable systems are not suitable for tasks that require the multirotor to maintain a certain distance from the payload, such as payload release and suspension missions involving human operators. Keeping the multirotor at a designated height while extending the cable length can enhance personnel safety. To address these limitations, some recent works have employed the variable-length cable for payload delivery, thereby significantly enhancing the system's flexibility. Specifically, the control objectives can be divided into two categories: one focusing on multirotor motion tracking (Huang et al., 2023; Liang et al., 2023; Yu et al., 2024, 2025) and the other on payload motion tracking (Zeng et al., 2019). By mounting a motor beneath the multirotor, the cable length can be dynamically adjusted during flight, facilitating smooth navigation through narrow spaces and enabling sample collection tasks, as demonstrated by Liang et al. (2023). Furthermore, an adaptive control scheme is designed for the aerial transportation system to realize payload landing on mobile platforms by eliminating payload swing in Yu et al. (2024), and a visual-based servoing control method is proposed in Yu et al. (2025) to improve the autonomy of payload transportation. To expedite the convergence of payload swing angles, Huang et al. (2023) introduces an enhanced coupling signal enabling the multirotor to track a desired point.

In summary, these efforts primarily focus on the motion tracking of the multirotor and the cable length, with particular emphasis on suppressing payload oscillations. While specific operations, such as the precise release of payloads at designated target points, necessitate direct manipulation of the payload. The traditional method (Yu et al., 2024) that indirectly controls payload position by suppressing its oscillations through adjustments to the multirotor and cable length exhibits limitations in task execution efficiency. To achieve direct payload position tracking, a geometric control scheme is designed in Zeng et al. (2019), which integrates the payload position and cable length into a single control unit. However, this approach requires predefined cable length trajectories, thus restricting the system's flexibility in task execution. For aerial transportation system with variable-length cable, the position of the payload is determined not only by the multirotor's motion but also by the length of the suspension cable. Actually, the same payload trajectory can be accomplished using different combinations of multirotor and cable length motions. Once the payload trajectory is established, it is necessary to consider the constraints related to multirotor and cable length motion tailored to the specific requirements, to identify the optimal motion combination. For instance, during the payload pickup and release phases, priority is given to extending the cable length

to mitigate ground effect and ensure personnel safety. Conversely, during long-distance transportation phases, the focus shifts to multirotor movement to guarantee efficient transportation. Currently, the coordination of multirotor motion with cable length variations to achieve precise payload trajectory tracking remains an open area of research.

To address these challenges, this paper presents a backstepping controller accompanied by an online cable length generator for the aerial transportation system. The developed control scheme is capable of achieving desired payload position tracking while simultaneously adjusting the cable length. The generator is specifically designed to balance the variations in cable length with multirotor movements. The main contributions of this paper can be summarized as follows:

- (1) Although the variable cable length increases the flexibility of aerial transportation systems, it also intensifies the nonlinearity and complex dynamics among the multirotor, cable and payload. By employing the backstepping procedure, a cascaded control scheme composed of four parts is designed to achieve payload trajectory tracking control, cable length and direction control, and multirotor attitude control. Without any linearization, the asymptotic stability of the closed-loop system is guaranteed through Lyapunov techniques and the growth restriction condition.
- (2) After the payload trajectory is given, a cable length generator is designed to manage the motion of the multirotor and cable length based on state constraints, thereby eliminating the need for manually predefined cable trajectories. Notably, the developed control law is incorporated into the system model to determine the dynamic constraints of the generator, enabling real-time operation.

The remainder of this paper is organized as follows: Section 2 presents the system modeling and problem statements. The controller design process is described in Section 3, followed by the cable length generator in Section 4. Section 5 provides the stability analysis, and Section 6 presents simulation results to demonstrate the performance of the proposed scheme. Finally, Section 7 concludes the paper and outlines directions for future work.

## 2. Problem statement

### 2.1. Notations

In this paper, vectors are distinguished by using bold letters. For  $\mathbf{a} \in \mathbb{R}^3$ , the hat map  $\hat{\cdot} : \mathbb{R}^3 \rightarrow \mathfrak{so}(3)$  is defined by  $\hat{\mathbf{a}}\mathbf{b} = \mathbf{a} \times \mathbf{b}$  for all  $\mathbf{b} \in \mathbb{R}^3$ ; the vee map  $(\cdot)^\vee : \mathfrak{so}(3) \rightarrow \mathbb{R}^3$  is its inverse. The 2-sphere is given by  $S^2 := \{\mathbf{p} \in \mathbb{R}^3 \mid \mathbf{p}^\top \mathbf{p} = 1\}$ . For a vector  $\mathbf{h} \in \mathbb{R}^n$ , define  $\text{Cosh}(\mathbf{h}) = [\cosh(h_1), \dots, \cosh(h_n)]^\top$ ,  $\text{Tanh}(\mathbf{h}) = [\tanh(h_1), \dots, \tanh(h_n)]^\top$ ,  $\text{Sech}(\mathbf{h}) = [\text{sech}(h_1), \dots, \text{sech}(h_n)]^\top$ ,  $\text{Ln}(\mathbf{h}) = [\ln(h_1), \dots, \ln(h_n)]^\top$ , and  $\text{diag}(\mathbf{h})$  is the diagonal matrix with diagonal elements of  $h_1, \dots, h_n$ .  $\lambda_{\max}(\cdot)$  and  $\lambda_{\min}(\cdot)$  denote the maximum eigenvalue and minimum eigenvalue of a matrix, respectively. For any vectors  $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^3$ , the following triple product identities are satisfied:  $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a}^\top \mathbf{c}) - \mathbf{c}(\mathbf{a}^\top \mathbf{b})$ ,  $\mathbf{a}^\top (\mathbf{b} \times \mathbf{c}) = \mathbf{b}^\top (\mathbf{c} \times \mathbf{a}) = \mathbf{c}^\top (\mathbf{a} \times \mathbf{b})$ . For a unit vector  $\mathbf{p} \in \mathbb{R}^3$  and an arbitrary vector  $\mathbf{a} \in \mathbb{R}^3$ , when they are orthogonal to each other, i.e.,  $\mathbf{p}^\top \mathbf{a} = 0$ , one has  $-\hat{\mathbf{p}}^2 \mathbf{a} = \mathbf{a}$ .

### 2.2. Dynamic model

Fig. 1 provides the schematic diagram of the aerial transportation system with variable-length cable, where  $\mathcal{I}$  denotes the right-hand inertia frame and  $\mathcal{B}$  represents the body-fixed frame. The symbols and definitions utilized throughout the paper are

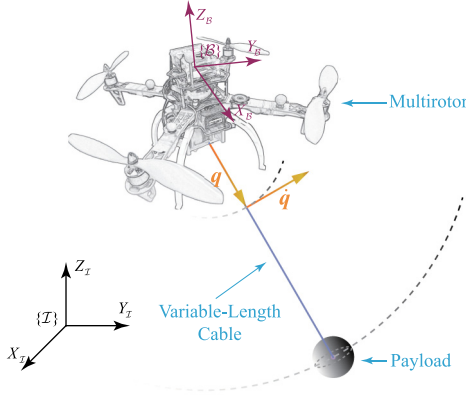


Fig. 1. Schematic of the aerial transportation system with variable-length cable.

**Table 1**  
Symbols and definitions of the system.

| Symbol                                        | Definition                                      |
|-----------------------------------------------|-------------------------------------------------|
| $m_Q, m_L \in \mathbb{R}$                     | Mass of multirotor and payload                  |
| $g \in \mathbb{R}$                            | Gravitational acceleration                      |
| $L \in \mathbb{R}$                            | Cable length                                    |
| $J \in \mathbb{R}^{3 \times 3}$               | Multirotor's moment of inertia                  |
| $R \in SO(3)$                                 | Rotation matrix from $\{B\}$ to $\{Z\}$         |
| $\mathbf{x}_L, \mathbf{v}_L \in \mathbb{R}^3$ | Payload position and velocity                   |
| $\mathbf{x}_Q, \mathbf{v}_Q \in \mathbb{R}^3$ | Multirotor position and velocity                |
| $\mathbf{q} \in S^2$                          | Direction of the cable                          |
| $\boldsymbol{\omega} \in \mathbb{R}^3$        | Angular velocity of the cable                   |
| $\boldsymbol{\Omega} \in \mathbb{R}^3$        | Angular velocity of multirotor in frame $\{B\}$ |
| $f_L \in \mathbb{R}$                          | Payload hoisting/lowering force                 |
| $f \in \mathbb{R}$                            | Applied thrust generated by multirotor          |
| $\boldsymbol{\tau} \in \mathbb{R}^3$          | Torque generated by multirotor                  |
| $\mathbf{e}_3 = [0, 0, 1]^T \in \mathbb{R}^3$ | Unit vector along z-axis                        |

summarized in Table 1. Unlike previous works, the cable length is modeled as a system state, rather than a constant, and can be actively extended or retracted via motor-driven mechanisms. Therefore, the control inputs of the system comprise not only the thrust  $f \in \mathbb{R}$  and torque  $\boldsymbol{\tau} \in \mathbb{R}^3$  generated by the multirotor, but also the force  $f_L \in \mathbb{R}$  for hoisting and lowering the payload produced by the cable-length regulation mechanism. The kinematics are described by  $\dot{\mathbf{x}}_L = \mathbf{v}_L$ ,  $\dot{\mathbf{q}} = \boldsymbol{\omega} \times \mathbf{q}$ , and  $\dot{R} = R\boldsymbol{\Omega}$ . Employing the angular momentum theorem (Hinrichsen & Pritchard, 2005) and Newton–Euler equation (Spong et al., 2020), the dynamic model can be established:

$$m_L(\dot{\mathbf{v}}_L + g\mathbf{e}_3) = f_L\mathbf{q}, \quad (1)$$

$$\frac{m_Q + m_L}{m_L}f_L - m_Q\ddot{L} = \mathbf{q}^T f \mathbf{R} \mathbf{e}_3 - m_Q L (\dot{\mathbf{q}}^T \dot{\mathbf{q}}), \quad (2)$$

$$m_Q L \dot{\boldsymbol{\omega}} = -\mathbf{q} \times f \mathbf{R} \mathbf{e}_3 - 2m_Q \dot{L} \boldsymbol{\omega}, \quad (3)$$

$$J \dot{\boldsymbol{\Omega}} = \boldsymbol{\tau} - \boldsymbol{\Omega} \times J \boldsymbol{\Omega}. \quad (4)$$

The full derivation is provided in Yu et al. (2025). Since this paper focuses on payload trajectory tracking, the model is constructed based on the payload's position and velocity. As illustrated in Fig. 1, the relationship between the multirotor and the payload can be described by  $\mathbf{x}_Q = \mathbf{x}_L - L\mathbf{q}$ . Thus, by performing a variable substitution, the dynamics can be reformulated in terms of the multirotor's position and velocity.

### 2.3. Control objective and error dynamics

For the convenience of subsequent control scheme design and analysis, the state errors of the system are first defined as follows. The payload position and velocity errors are given by:  $\mathbf{e}_x :=$

$\mathbf{x}_L - \mathbf{x}_{Ld}$ ,  $\mathbf{e}_v := \mathbf{v}_L - \dot{\mathbf{x}}_{Ld}$ , where  $\mathbf{x}_{Ld} \in \mathbb{R}^3$  represents the desired payload trajectory. The cable direction and angular velocity errors are defined as:  $\mathbf{e}_q := \mathbf{q}_d \times \mathbf{q}$ ,  $\mathbf{e}_\omega := \boldsymbol{\omega} + \dot{\mathbf{q}}^T \boldsymbol{\omega}_d$ , where  $\mathbf{q}_d \in S^2$  and  $\boldsymbol{\omega}_d := \mathbf{q}_d \times \dot{\mathbf{q}}_d \in \mathbb{R}^3$  denote the desired cable direction and angular velocity, respectively. The cable length error is defined as:  $e_L := L - L_d$ , where  $L_d \in \mathbb{R}$  is the desired cable length. The multirotor attitude and angular velocity errors are defined as:  $\mathbf{e}_R := \frac{1}{2}(R_d^T R - R^T R_d)^V$ ,  $\mathbf{e}_\Omega := \boldsymbol{\Omega} - R^T R_d \boldsymbol{\Omega}_d$ , where  $R_d \in SO(3)$  is the desired rotation matrix and  $\boldsymbol{\Omega}_d \in \mathbb{R}^3$  is the desired multirotor angular velocity.

The control objective is to design a feedback control law that drives the payload to its desired trajectory, while simultaneously adjusting the cable length and direction to achieve their respective desired states. This can be mathematically described as follows:

$$\mathbf{e}_x \rightarrow \mathbf{0}_{3 \times 1}, \mathbf{e}_v \rightarrow \mathbf{0}_{3 \times 1}, \mathbf{e}_q \rightarrow \mathbf{0}_{3 \times 1}, \mathbf{e}_\omega \rightarrow \mathbf{0}_{3 \times 1},$$

$$e_L \rightarrow 0, \mathbf{e}_R \rightarrow \mathbf{0}_{3 \times 1}, \mathbf{e}_\Omega \rightarrow \mathbf{0}_{3 \times 1}.$$

Subsequently, by substituting the system errors into the system dynamics model (1)–(4), the open-loop error dynamics of the system can be rearranged as:

$$m_L \dot{\mathbf{e}}_v = f_L \mathbf{q} - m_L \ddot{\mathbf{x}}_{Ld} - m_L g \mathbf{e}_3, \quad (5)$$

$$m_Q \ddot{e}_L = -\mathbf{q}^T f \mathbf{R} \mathbf{e}_3 + \frac{m_Q + m_L}{m_L} f_L + m_Q L (\dot{\mathbf{q}}^T \dot{\mathbf{q}}) - m_Q \ddot{L}_d, \quad (6)$$

$$m_Q L \dot{\mathbf{e}}_\omega = -\mathbf{q} \times f \mathbf{R} \mathbf{e}_3 - 2m_Q \dot{L} \boldsymbol{\omega} + m_Q L (\mathbf{q}^T \boldsymbol{\omega}_d) \dot{\mathbf{q}} + m_Q L (\boldsymbol{\omega}_d^T \dot{\mathbf{q}} + \mathbf{q}^T \dot{\boldsymbol{\omega}}_d) \mathbf{q} - m_Q L \dot{\boldsymbol{\omega}}_d, \quad (7)$$

$$J \dot{\mathbf{e}}_\Omega = \boldsymbol{\tau} - \boldsymbol{\Omega} \times J \boldsymbol{\Omega} + J (\hat{\boldsymbol{\Omega}} R^T R_d \boldsymbol{\Omega}_d - R^T R_d \dot{\boldsymbol{\Omega}}_d). \quad (8)$$

### 3. Controller design

In this section, the design process of the payload trajectory tracking controller is presented through the backstepping procedure. The control scheme is developed from four aspects: payload position control, cable length control, cable direction control, and multirotor attitude control. The block diagram of the proposed algorithm is depicted in Fig. 2. The pink blocks represent the desired payload trajectory and cable length trajectory generator modules. The green blocks denote the control modules. The purple blocks illustrate the calculations conducted between these modules. Finally, the blue block represents the aerial transportation system.

#### 3.1. Payload position control

The payload position control scheme is designed based on the error dynamics (5). According to the hierarchical control approach (Kendoul, 2009; Zhao et al., 2014), let  $f_L \mathbf{q} := \mathbf{F}_L + \Delta_L$ , where  $\mathbf{F}_L = \frac{f_L}{q_d^T \mathbf{q}} \mathbf{q}_d \in \mathbb{R}^3$  is the virtual control input to be constructed, and  $\Delta_L = \frac{f_L}{q_d^T \mathbf{q}} [(\mathbf{q}_d^T \mathbf{q}) \mathbf{q} - \mathbf{q}_d] \in \mathbb{R}^3$  is the coupling term between the virtual input  $\mathbf{F}_L \in \mathbb{R}^3$  and the cable direction error  $\mathbf{e}_q$ . As seen from the decomposition,  $\mathbf{F}_L$  and  $\Delta_L$  can be obtained by multiplying the same scalar  $\frac{f_L}{q_d^T \mathbf{q}}$  with the vectors  $\mathbf{q}_d$  and  $(\mathbf{q}_d^T \mathbf{q}) \mathbf{q} - \mathbf{q}_d$ , respectively. Consequently, the geometric relationship between  $\mathbf{q}_d$  and  $(\mathbf{q}_d^T \mathbf{q}) \mathbf{q} - \mathbf{q}_d$  in Fig. 3 is directly analogous to that between  $\mathbf{F}_L$  and  $\Delta_L$ . Using the vector triple-product identity,  $\Delta_L$  can be further expressed as:  $\Delta_L = -\frac{f_L}{q_d^T \mathbf{q}} \mathbf{q} \times (\mathbf{q}_d \times \mathbf{q}) = -\frac{f_L}{q_d^T \mathbf{q}} \mathbf{q} \times \mathbf{e}_q$ . Taking norms and recalling  $\|\mathbf{F}_L\| = \frac{|f_L|}{|q_d^T \mathbf{q}|} \|\mathbf{q}_d\| = \frac{|f_L|}{|q_d^T \mathbf{q}|}$  yields  $\|\Delta_L\| = \|\mathbf{F}_L\| \|\mathbf{q} \times \mathbf{e}_q\| \leq \|\mathbf{F}_L\| \|\mathbf{e}_q\|.$  (9)

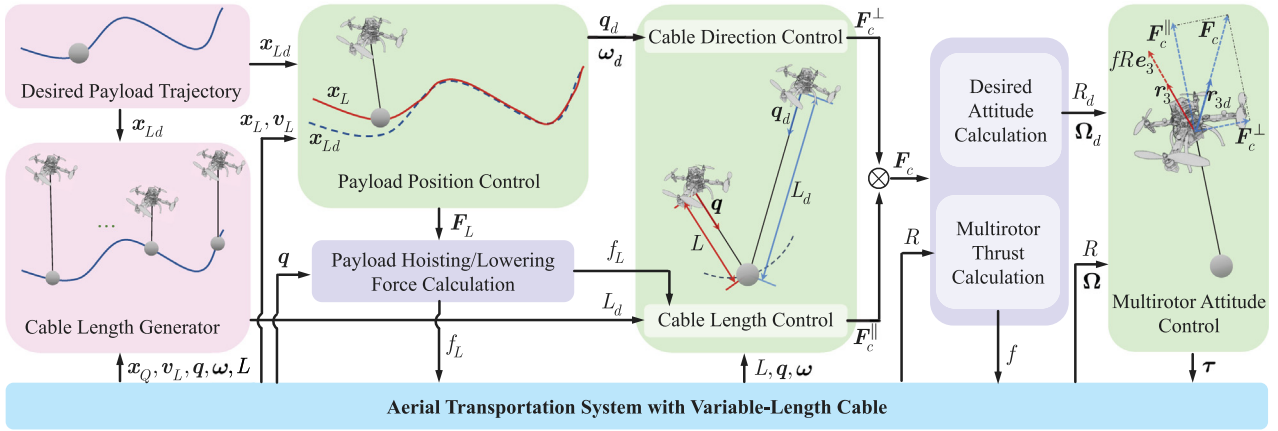
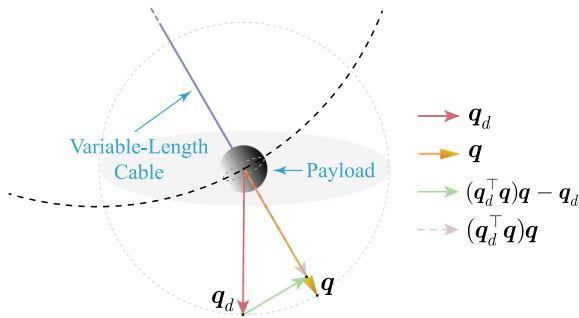


Fig. 2. Block diagram of the designed payload trajectory tracking algorithm.

Fig. 3. Geometric illustration of  $q$ ,  $q_d$ , and  $(q_d^\top q)q - q_d$ .

The payload position control scheme  $F_L$  is designed as:

$$F_L = -K_p \text{Tanh}(e_x) - K_d \text{Tanh}(e_v) + m_L \ddot{x}_{Ld} + m_L g e_3, \quad (10)$$

where diagonal matrices  $K_p = \text{diag}([k_{px}, k_{py}, k_{pz}])$  and  $K_d = \text{diag}([k_{dx}, k_{dy}, k_{dz}]) \in \mathbb{R}_+^{3 \times 3}$  are positive-definite gain matrices. Subsequently, the desired cable direction  $q_d$  can be obtained by  $q_d = -\frac{F_L}{\|F_L\|}$ . Thus, according to the decomposition of  $f_L q$ , one has  $F_L = \frac{f_L}{q_d^\top q} q_d = -\frac{f_L \|F_L\|}{F_L^\top q} \frac{F_L}{\|F_L\|} = \frac{f_L}{F_L^\top q} F_L$ , which indicates that the payload hoisting/lowering force  $f_L$  can be derived by  $f_L = F_L^\top q$ .

### 3.2. Cable length control

From the error dynamic model (6) and (7), it is evident that the term  $f R e_3$  impacts both the length and direction dynamics of the cable. Specifically, the component of  $f R e_3$  along  $q$  governs the cable length, whereas its component perpendicular to  $q$  dictates the direction of the cable. Define the virtual control input of  $f R e_3$  as  $F_c$ . To decouple the cable length and direction control,  $F_c$  is divided into two parts that satisfy the following equation:

$$F_c := F_c^\parallel + F_c^\perp, \quad (11)$$

where  $F_c^\parallel$  denotes the projection of  $F_c$  along  $q$ , and  $F_c^\perp$  is perpendicular to  $F_c^\parallel$ . Thereafter, the control design for the cable length will be addressed. According to the above analysis and the error dynamics (6), the cable length control law  $f_c := -q^\top F_c^\parallel$  can be constructed as:

$$f_c = -k_{pl} e_L - k_{dl} \dot{e}_L - \frac{m_Q + m_L}{m_L} f_L - m_Q L (\dot{q}^\top \dot{q}) + m_Q \ddot{L}_d - \frac{k_a e_L}{l^2 - e_L^2}, \quad (12)$$

where  $k_{pl}$ ,  $k_{dl}$  and  $k_a \in \mathbb{R}_+$  are positive control gains,  $l \in \mathbb{R}_+$  is a positive constant that specifies the admissible upper bound on the cable-length error.

### 3.3. Cable direction control

Based on the analysis of (11) and the error dynamics (7), the cable direction control law  $F_c^\perp$  can be designed as:

$$F_c^\perp = m_Q L \cdot q \times \left( -k_q e_q - k_\omega e_\omega - (q^\top \omega_d) \dot{q} - \hat{q}^2 \dot{\omega}_d + 2 \frac{\dot{L}}{L} \omega - \frac{k_b e_q}{\varrho^2 - \Psi_q} \right), \quad (13)$$

where  $\Psi_q := 1 - q^\top q_d$  is the configuration error for the  $S^2$  manifold,  $k_q$ ,  $k_\omega$  and  $k_b \in \mathbb{R}_+$  are positive control gains,  $\varrho \in (0, 1)$  is a positive constant that specifies the admissible upper bound on the cable-direction misalignment. Thus, according to (11), by collecting (12) and (13), the applied multirotor control thrust input can be obtained as  $f = F_c^\top R e_3$ .

### 3.4. Multirotor attitude control

The desired multirotor attitude can be defined as:  $R_d := [r_{1d}; r_{2d}; r_{3d}]$ , where  $r_{3d} \in S^2$  is calculated by  $r_{3d} = \frac{F_c}{\|F_c\|}$ ,  $r_{2d} = \frac{r_{3d} \times r_{1a}}{\|r_{3d} \times r_{1a}\|}$ , and  $r_{1d} = r_{2d} \times r_{3d}$ .  $r_{1a}$  is an arbitrary vector that is not parallel to  $r_{3d}$ . The desired angular velocity can be calculated as  $\Omega_d = (R_d^\top \dot{R}_d)^\vee$ . Following the approach similar to that in Sreenath et al. (2013), the multirotor attitude control is implemented as:

$$\tau = -\frac{1}{\epsilon^2} k_R e_R - \frac{1}{\epsilon} k_\Omega e_\Omega + \Omega \times J \Omega - J(\hat{\Omega} R^\top R_d \Omega_d - R^\top R_d \hat{\Omega}_d), \quad (14)$$

where  $k_R$ ,  $k_\Omega \in \mathbb{R}_+$  are positive control gains, and  $0 < \epsilon < 1$ . Through the singular perturbation theory (Sreenath et al., 2013), the parameter  $\epsilon$  is introduced to enable rapid exponential convergence of  $e_R$  and  $e_\Omega$ .

From the above subsections, one can observe that the desired cable direction  $q_d$ , cable angular velocity  $\omega_d$ , rotation matrix  $R_d$ , and multirotor angular velocity  $\Omega_d$  depend on the virtual control inputs  $F_L$  and  $F_c$  and their time derivatives. Fortunately, these quantities can be computed from the closed-loop system dynamics via systematic derivations, without requiring acceleration or higher-order derivatives of the states. Consequently, the proposed method is readily deployable on real systems, with no need for numerical differentiation of the states.



#### 4. Cable length generator

This section develops the cable length generator. According to the aforementioned backstepping control schemes, the cable length trajectory is set to be fourth-order continuous to ensure the smoothness of the control input. A generalized state vector  $\xi$  is introduced to describe the nonlinear optimization problem, i.e.,  $\xi := (\mathbf{x}_Q, \mathbf{q}, L, \mathbf{v}_Q, \boldsymbol{\omega}, \dot{L}, \mathbf{L}_t) \in \mathbb{R}^3 \times S^2 \times \mathbb{R} \times \mathbb{R}^3 \times \mathbb{R}^3 \times \mathbb{R} \times \mathbb{R}^5$ , where  $\mathbf{L}_t := [L_d, \dot{L}_d, \ddot{L}_d, L_d^{(3)}, L_d^{(4)}]^\top \in \mathbb{R}^5$  is the generalized desired cable length trajectory including  $L_d$  as well as its first to fourth time derivatives.  $\mathbf{X}_d := (\mathbf{x}_{Ld}, \dot{\mathbf{x}}_{Ld}, \ddot{\mathbf{x}}_{Ld}, \mathbf{x}_{Ld}^{(3)}, \mathbf{x}_{Ld}^{(4)}) \in \mathbb{R}^{15}$  is the generalized desired trajectory vector of the payload. To obtain a reasonable cable length trajectory, the general form of the cable length generator is designed as follows:

$$\min_{L_d^{(5)}} \int_t^{t+T} \ell(\xi(\delta), L_d^{(5)}(\delta)) d\delta \quad (15)$$

$$\text{s.t. } \dot{\xi} = f_\xi(\xi, \mathbf{X}_d, L_d^{(5)}), \quad (16)$$

$$\underline{\mathbf{L}}_t \leq \mathbf{L}_t \leq \bar{\mathbf{L}}_t, \quad (17)$$

$$\underline{L}_d^{(5)} \leq L_d^{(5)} \leq \bar{L}_d^{(5)}, \quad (18)$$

where (15) is the cost function, (16) is the generalized dynamic constraint, (17) is the constraint of the generalized desired cable length, and (18) is the constraint of the output. The cable length generator is model-based and controller-aware. Under the proposed backstepping control schemes, the cable-length trajectory must be  $C^4$  continuous to ensure smooth control inputs. Accordingly, the fifth derivative of the desired cable length  $L_d^{(5)}$ , is chosen as the optimization output. Since the cable-length generator is devoted to coordinating the multirotor motion with cable-length variations, a generalized dynamical system (16) is adopted as the dynamic constraint of the generator. Its state vector comprises the multirotor motion, the cable length and its direction, and the desired cable length along with its first-fourth derivatives. According to the relationship between the multirotor and the payload (1), the dynamic constraint (16) can be obtained by substituting the payload position control scheme (10), along with the cable length and direction control laws (12), (13) into the dynamic model (1), (2) and (3). (17) is the constraint of the generalized desired cable length, where  $\underline{\mathbf{L}}_t, \bar{\mathbf{L}}_t \in \mathbb{R}^5$  are the lower and upper bounds, respectively. Constraint (18) governs the output, specifying the lower bound  $\underline{L}_d^{(5)}$  and the upper bound  $\bar{L}_d^{(5)}$ . Tailored to different task requirements, the cost function  $\ell$  is designed to enhance coordination between the multirotor's motion and cable length variations, integrating this with the system's state.

#### 5. Stability analysis

The stability of the closed-loop system is analyzed in this section. By neglecting the coupling term  $\Delta_L$ , the stability of the payload position, cable length and direction are first proven individually. Subsequently, based on the theory of cascade systems, the stability of the overall system is ensured by demonstrating that the coupling term satisfies the growth restriction condition (Kendoul, 2009; Zhao et al., 2014).

**Theorem 1.** *The designed payload position control law (10) guarantees the convergence of the payload position and velocity errors to zero asymptotically, i.e.,*

$$\lim_{t \rightarrow \infty} \mathbf{e}_x = \mathbf{0}_{3 \times 1}, \lim_{t \rightarrow \infty} \mathbf{e}_v = \mathbf{0}_{3 \times 1}.$$

**Proof.** Please refer to Yu et al. (2025).

**Theorem 2.** *Consider the closed-loop system under the control laws (12) and (13). Let the barrier Lyapunov functions be defined on the safe sets  $S_L := \{|e_L| < \iota\}$  and  $S_q := \{\Psi_q < \varrho^2\}$ , where  $\iota \in \mathbb{R}_+$  and  $\varrho \in (0, 1)$  are positive constants. If the initial errors satisfy  $|e_L(0)| < \iota$  and  $\Psi_q(0) < \varrho^2$ , then there exist positive control parameters  $k_{pl}, k_{dl}, k_a, k_q, k_\omega, k_b \in \mathbb{R}_+$  such that:*

(1)  $|e_L(t)| < \iota$  and  $\Psi_q(t) < \varrho^2$  for all  $t \geq 0$ ;

(2) the cable length and direction tracking errors converge exponentially to zero, i.e.,

$$\lim_{t \rightarrow \infty} e_L = 0, \lim_{t \rightarrow \infty} \dot{e}_L = 0, \lim_{t \rightarrow \infty} \mathbf{e}_q = \mathbf{0}_{3 \times 1}, \lim_{t \rightarrow \infty} \mathbf{e}_\omega = \mathbf{0}_{3 \times 1}.$$

**Proof.** Please refer to Yu et al. (2025).

**Theorem 3.** *The proposed control law (10), (12), (13) and (14) can drive the payload to the desired trajectory, and adjust the cable length and direction to the desired ones, implying the following result:*

$$\lim_{t \rightarrow \infty} [\mathbf{e}_x^\top, \mathbf{e}_v^\top, e_L, \dot{e}_L, \mathbf{e}_q^\top, \mathbf{e}_\omega^\top, \mathbf{e}_R^\top, \mathbf{e}_\Omega^\top]^\top = \mathbf{0}_{20 \times 1}.$$

**Proof.** The stability analysis of the overall closed-loop system with the consideration of the coupling term  $\Delta_L$  is taken into account here. For more details, please refer to Yu et al. (2025).

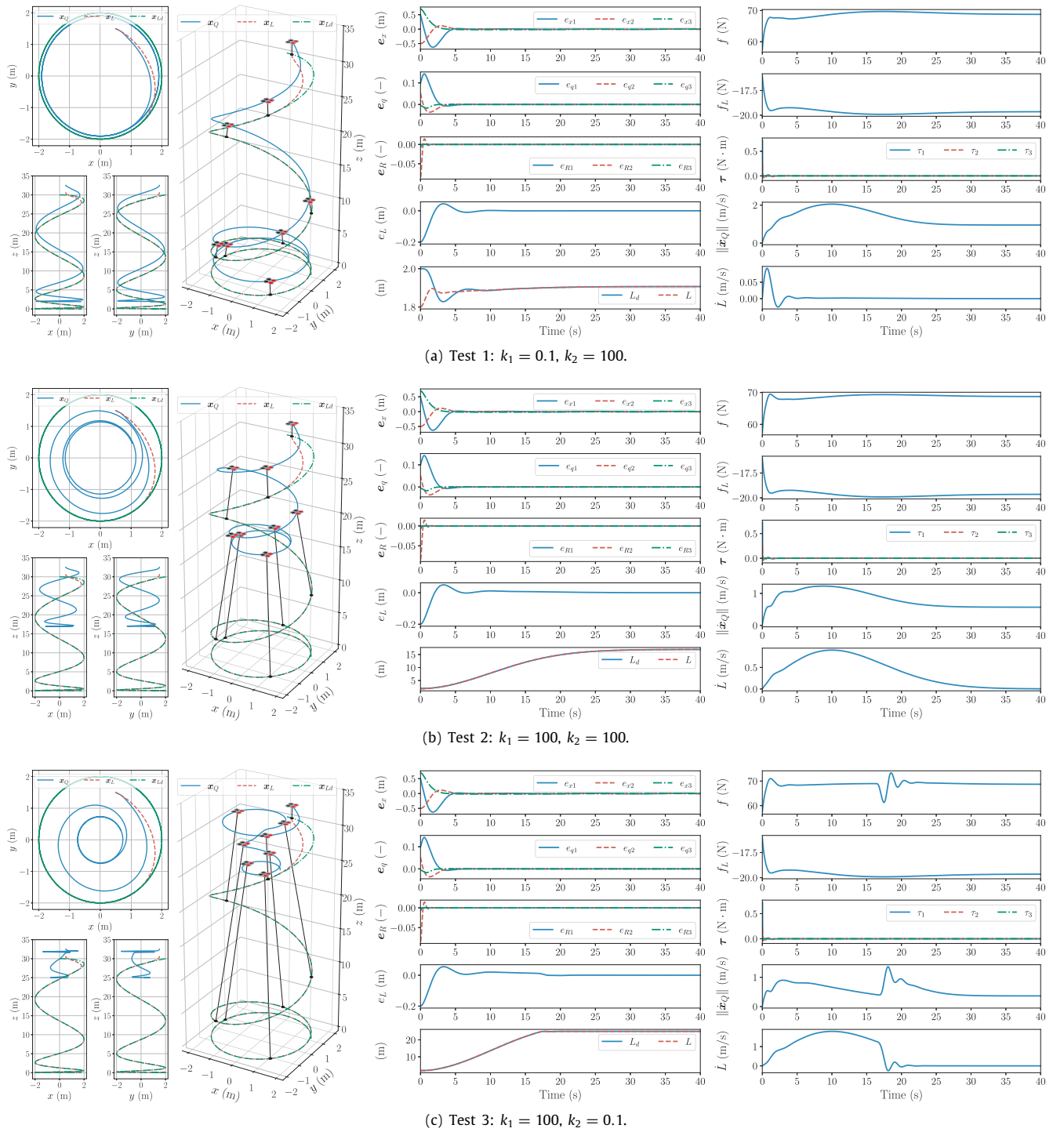
#### 6. Simulation study

To demonstrate the potential of the solution, several groups of simulation tests are implemented with the *acados* library (Verschuere et al., 2022) in this section. All simulations are run on a desktop computer equipped with an AMD Ryzen 9 9950X CPU and 64 GB of RAM. The control frequency is 100 Hz, corresponding to a controller sampling period of  $T_s = 10$  ms. The physical parameters are selected as  $m_Q = 5.0$  kg,  $m_L = 2.0$  kg,  $g = 9.81$  m/s<sup>2</sup>,  $J = \text{diag}([0.05, 0.05, 0.06])$  kg · m<sup>2</sup>. The control gains are chosen as  $K_p = \text{diag}([4, 4, 5])$ ,  $K_d = \text{diag}([4.5, 4.5, 6])$ ,  $k_{pl} = 4$ ,  $k_{dl} = 4.5$ ,  $k_q = 1.8$ ,  $k_\omega = 1.9$ ,  $k_a = 0.5$ ,  $k_b = 0.5$ ,  $\iota = 0.25$ ,  $\varrho = \sqrt{0.1}$ ,  $\frac{1}{\epsilon^2} k_R = 1.92$ ,  $\frac{1}{\epsilon} k_\Omega = 0.3$ . For additional simulation results and discussions related to this work, please refer to Yu et al. (2025).

##### 6.1. Simulation 1: Basic performance validation

The initial position of the payload is set as  $\mathbf{x}_L(0) = [0.5, 1.5, 32.5]^\top$  m, the initial direction of the cable is  $\mathbf{q}(0) = [0, 0, -1]^\top$ , and the initial cable length is  $L(0) = 1.8$  m. The reference payload trajectory is chosen as  $\mathbf{x}_{Ld} = [2 \sin(0.5t), 2 \cos(0.5t), 30 \exp(-0.005t^2)]^\top$  m. In this subsection, we focus on the basic performance validation, and the function  $\ell$  is selected to manage the motion of the multirotor with the variation of the cable-length, specifically as:  $\ell = k_1 \|\dot{\mathbf{x}}_Q\|^2 + k_2 \dot{L}^2 + \mathbf{L}_t^\top K \mathbf{L}_t + k_8 L_d^{(5)2}$ , where  $K = \text{diag}([k_3, k_4, k_5, k_6, k_7])$  and  $k_i, i = 1, \dots, 8$  are the weighting factors. Three tests are conducted by choosing different weighting factors  $k_1$  and  $k_2$ , i.e., (1) Test 1:  $k_1 = 0.1$ ,  $k_2 = 100$ ; (2) Test 2:  $k_1 = 100$ ,  $k_2 = 100$ ; (3) Test 3:  $k_1 = 100$ ,  $k_2 = 0.1$ . The rest weighting factors are set the same, i.e.,  $K = \text{diag}([0, 1.6, 3.2, 2.4, 0.8])$ ,  $k_8 = 0.1$ . The upper and lower bounds of  $\mathbf{L}_t$  and  $L_d^{(5)}$  are set as  $\underline{\mathbf{L}}_t = [0.5, -30, -30, -30, -30]^\top$ ,  $\bar{\mathbf{L}}_t = [25, 30, 30, 30, 30]^\top$ ,  $\underline{L}_d^{(5)} = -50$ ,  $\bar{L}_d^{(5)} = 50$ .

The simulation results are depicted in Fig. 4. The first and second columns illustrate the two-dimensional and three-dimensional trajectories of the payload and the multirotor, respectively. Analysis of the three tests reveals that when the weight of the cable length velocity in the cost function is high,



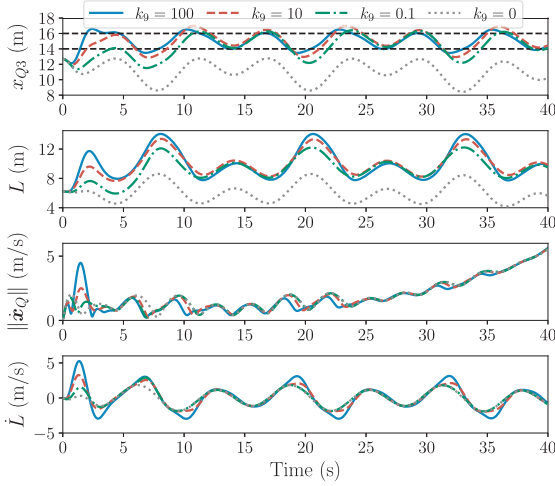
**Fig. 4.** Results for Simulation 1. (The first column presents the two-dimensional trajectories of the payload and the multirotor, while the second column displays their three-dimensional trajectories. The third column depicts the tracking errors for the payload's position, cable direction and length, and multirotor attitude, as well as the generated desired and actual cable length curves. The fourth column shows the multirotor's thrust force and torque, the payload's hoisting/lowering force, and the velocities of both the multirotor and the cable length.)

there is minimal change in the cable length, effectively maintaining a fixed length to achieve precise trajectory tracking of the payload. Conversely, when the weight of the multirotor velocity is increased, the multirotor exhibits reduced motion variability, and the system compensates by extending the cable length to ensure accurate payload trajectory tracking. Notably, as demonstrated in Test 3, when constraints are triggered, i.e., the cable reaching its maximum length, the payload trajectory tracking is further

achieved by reducing the multirotor's altitude. The third column presents the tracking errors for payload position, cable direction and length, and multirotor attitude, along with the generated desired and actual cable length curves. Despite initial nonzero errors, all error signals converge to zero within approximately 5 s, validating the efficacy of the proposed control method in enabling the aerial transportation system with variable-length cable to accurately follow the desired trajectory. The fourth column displays

**Table 2**  
acados per-step solve time of Simulation 1.

|        | mean (ms) | max (ms) | std (ms) |
|--------|-----------|----------|----------|
| Test 1 | 5.114     | 7.710    | 0.065    |
| Test 2 | 4.931     | 6.104    | 0.073    |
| Test 3 | 5.228     | 7.753    | 0.487    |



**Fig. 5.** Results for Simulation 2 Test 1. (The multirotor altitude, cable length, multirotor velocity, and cable length velocity.)

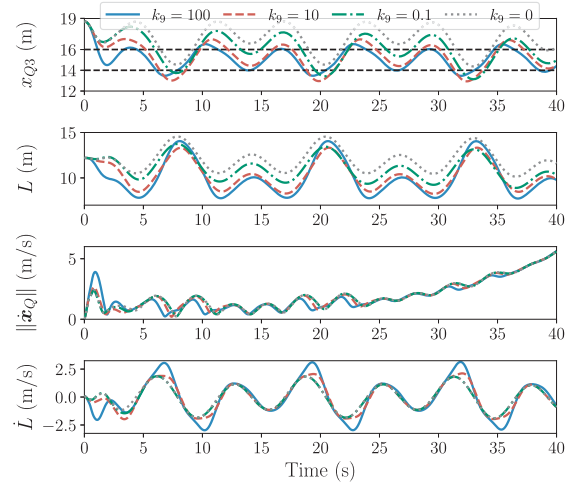
the multirotor thrust force and torque, payload hoisting/lowering force, and the velocities of both the multirotor and cable length. These curves distinctly illustrate how the multirotor's movement and the variation in cable length are influenced by different weighting factors. Table 2 reports per-step solve-time statistics including mean, maximum, and standard deviation (std) values. Across the three tests, the mean solve time is 4.931–5.228 ms (49.3%–52.3% of  $T_s$ ), with std 0.065–0.487 ms indicating low jitter overall and a larger variability in Test 3. The maximum solve times are 6.104 ms (Test 2), 7.710 ms (Test 1), and 7.753 ms (Test 3), corresponding to utilizations of 61.0%, 77.1%, and 77.5% of the sampling period  $T_s$ , respectively, thus leaving timing headroom of 3.896 ms, 2.290 ms, and 2.247 ms. These results indicate sufficient real-time margin on the stated hardware.

## 6.2. Simulation 2: Performance validation considering multirotor altitude

During the execution of transportation tasks, it is typically essential to maintain the multirotor within a specified safe altitude range. Therefore, to further assess the performance of the cable length generator, a soft constraint on multirotor's altitude is implemented, based on the cost function detailed in the previous subsection, specifically as follows:  $\ell = k_1 \|\dot{\mathbf{x}}_Q\|^2 + k_2 \dot{L}^2 + \mathbf{L}_t^T \mathbf{K} \mathbf{L}_t + k_8 L_d^{(5)2} + k_9 \ell_{\text{alt}}$ , where  $\ell_{\text{alt}} = \exp(k_z(z_Q - x_{Q3})) + \exp(k_z(x_{Q3} - \bar{z}_Q))$ ,  $z_Q$  and  $\bar{z}_Q$  are the expected lower and upper bounds of the multirotor's altitude, and  $k_z$  is a positive constant. The reference payload trajectory is chosen as

$$\mathbf{x}_{Ld} = \begin{bmatrix} \exp(0.1t) \\ \cos(0.5t) + \sin(0.3t + \frac{\pi}{2}) \\ 2 \sin(0.5t + \frac{\pi}{4}) + 3 \cos(t + \frac{\pi}{2}) + 5 \end{bmatrix} \text{ m.}$$

Two sets of tests are conducted by choosing different initial multirotor positions and cable lengths, i.e., (1) Test 1:  $\mathbf{x}_Q(0) =$



**Fig. 6.** Results for Simulation 2 Test 2. (The multirotor altitude, cable length, multirotor velocity, and cable length velocity.)

**Table 3**  
acados per-step solve time of Simulation 2.

|        | Value of $k_9$ | mean (ms) | max (ms) | std (ms) |
|--------|----------------|-----------|----------|----------|
| Test 1 | 100            | 4.878     | 7.605    | 0.197    |
|        | 10             | 4.902     | 6.446    | 0.050    |
|        | 0.1            | 4.922     | 6.187    | 0.059    |
|        | 0              | 4.895     | 6.472    | 0.135    |
| Test 2 | 100            | 4.682     | 6.513    | 0.093    |
|        | 10             | 5.126     | 6.652    | 0.152    |
|        | 0.1            | 4.913     | 5.989    | 0.141    |
|        | 0              | 4.935     | 5.799    | 0.038    |

$[0.9, 2.1, 12.7]^T$  m,  $L(0) = 6.2$  m; (2) Test 2:  $\mathbf{x}_Q(0) = [0.9, 2.1, 18.7]^T$  m,  $L(0) = 12.2$  m. The target altitude bounds for the multirotor are set between  $z_Q = 14$  and  $\bar{z}_Q = 16$ . For both sets of simulations, the initial cable direction is established as  $\mathbf{q}(0) = [0, 0, -1]^T$ . Weighting factors are chosen as  $k_1 = 100$ ,  $k_2 = 100$ ,  $\mathbf{K} = \text{diag}([0, 1.6, 3.2, 2.4, 0.8])$ ,  $k_8 = 0.1$ . The upper and lower bounds of  $\mathbf{L}_t$  and  $L_d^{(5)}$  are set as  $\mathbf{L}_t = [0.5, -60, -60, -60, -60]^T$ ,  $\bar{\mathbf{L}}_t = [55, 60, 60, 60, 60]^T$ ,  $L_d^{(5)} = -500$ ,  $\bar{L}_d^{(5)} = 500$ .

The simulation results are displayed in Figs. 5 and 6. In each set of simulation, the values of  $k_9$  are selected as 100, 10, 0.1, and 0 to verify the effectiveness of the cable length generator. The results indicate that the term  $\ell_{\text{alt}}$  effectively imposes a constraint on the altitude of the multirotor, with a larger value of  $k_9$  leading to a more pronounced constraint on the multirotor's altitude. By comparing Figs. 5 and 6, it can be seen that regardless of whether the multirotor's initial altitude is above or below the specified range, the incorporation of the cost  $\ell_{\text{alt}}$  enables the designed cable length generation scheme to effectively regulate the multirotor's altitude to approach the desired range. Table 3 summarizes the per-step acados solve time in Simulation 2 for four values of  $k_9$ . Across all runs, the mean solve time lies in 4.682–5.126 ms (46.8–51.3% of  $T_s$ ), while the maxima are 5.799–7.605 ms (58.0–76.1% of  $T_s$ ), leaving at least 2.395 ms of slack. These timings comfortably meet the 100 Hz real-time requirement on the stated hardware.

## 7. Conclusions

This paper explores a payload trajectory tracking scheme for the aerial transportation system with variable-length cable. The proposed scheme includes a backstepping controller for payload trajectory, cable length and direction tracking, as well as

an online cable length generator. The generator integrates the backstepping controller into the dynamic system model constraints and optimizes the desired cable length in real-time for the control scheme to follow. By employing Lyapunov techniques and growth restriction conditions, the proposed control scheme ensures the asymptotic stability of the closed-loop system. Under a reasonable cost function, the cable length generator effectively allocates the multirotor's motion and cable length, facilitating consistent payload trajectory tracking across various configurations of multirotor positions and cable lengths. Simulation results further highlight the advantages of the proposed control scheme, demonstrating its effectiveness in both trajectory tracking and cable length generation. In future work, we will investigate novel control schemes to strengthen the theoretical robustness guarantees of variable-length cable-suspended aerial transportation systems.

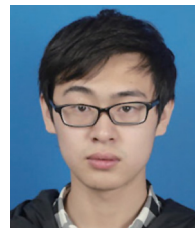
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